AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 10 · STUDENT LESSON

If/Then Decisions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Update. Test. Choose one path.

Trace current-state decisions, test equality, preserve unknown inputs and inspect a new brightness procedure.

Keep one habit: Name the test-point state → evaluate the exact comparison → execute its selected branch.

MAIN LESSON · SECTION 1

Which branch runs, and when?

> Track the current state, the equality case and the exact instructions after a decision.

Start with an updated value

A value can cross a threshold after an update. Test the value that exists when the decision is reached, then carry out the selected branch once.

Retrieve the current-value update

Score starts at 4. Add 2 to score, then double it. Which value is doubled, and what is the final score?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can evaluate a **condition** using the current state and trace only the selected **branch**.
- I can distinguish inclusive and strict boundaries using values below, at and above a **threshold**.
- I can design a **boundary test** and handle an unknown **input** without silently treating it as false.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

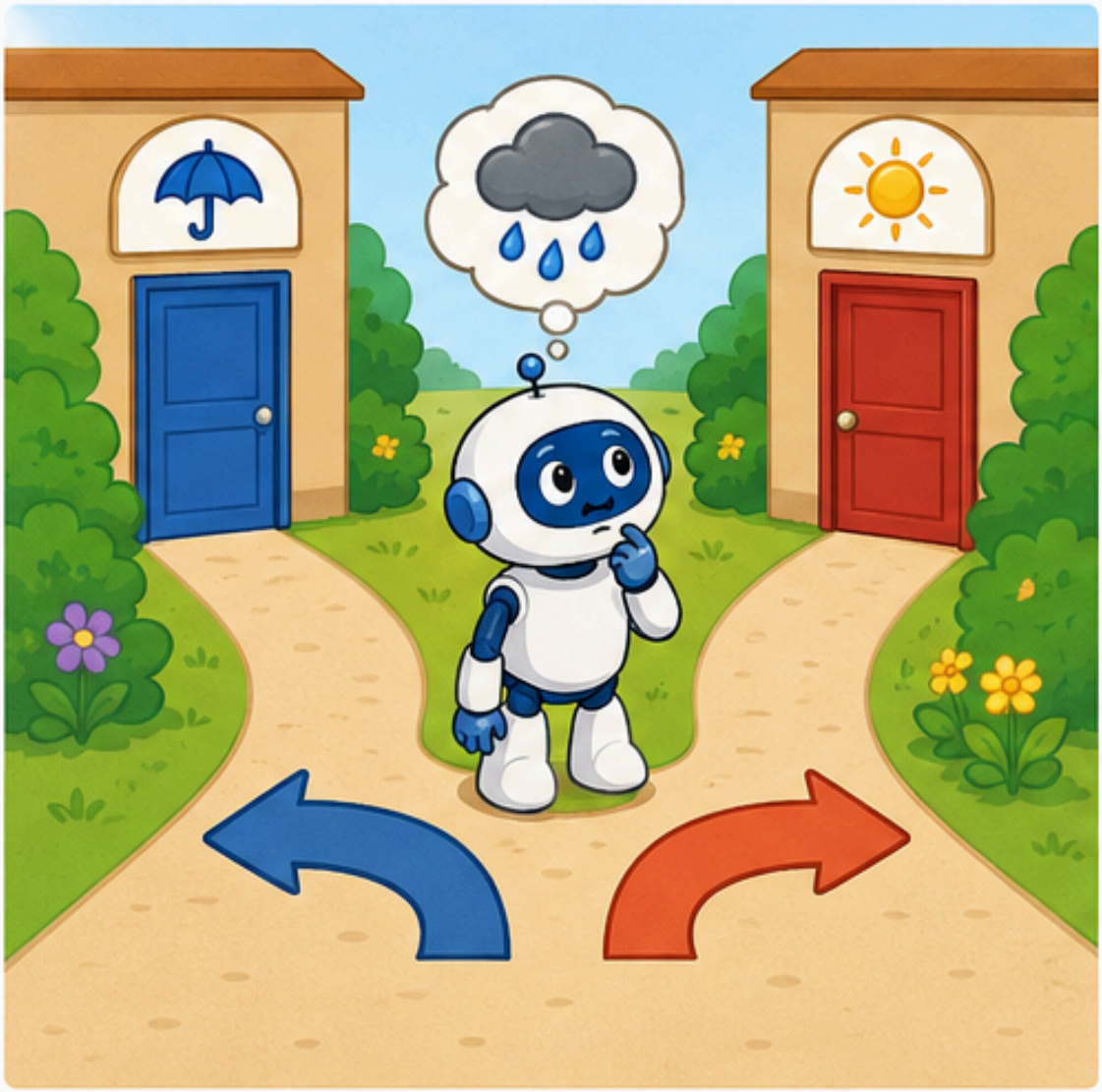
Remember before reading

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

Conditions make an algorithm **branch** when information changes.
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LEARNING TARGET A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can trace exactly one **branch** and test a boundary that includes equality.

Update first, then test the current score

> The paper game contains one if/else decision and an explicit report-and-stop instruction.

Read the complete paper-game recipe

A paper game updates a score with this procedure: add 2 to score; if score is at least 6, subtract 3 from score; otherwise add 4 to score; report score and stop. The if and otherwise instructions form one if/**else** decision.

Exactly one of those **branch**es runs each time this decision is reached.

Interpret “at least” before choosing

A **condition** asks whether a statement is true for the current state. At least 6 means 6 or any larger number. If score is 6, the **condition** is true. If score is 5, it is false.

The **condition** itself does not change score; the chosen **branch** performs the update.

Trace the start at 4

Start score at 4. First add 2, giving 6. Now test the current 6: the **condition** is true, so subtract 3 and report 3. Testing the original 4 would choose the wrong **branch**.

Write the state immediately before the **condition** so the **evidence** for the choice is visible.

Explain the selected branch

- Start at 4 and mark the value immediately before the condition. Does testing the condition itself change that value? Which one branch runs, and where does the procedure go next?

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY A clothing helper follows different paths depending on temperature, rain, and wind **condition**s.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

MAIN LESSON · SECTION 3

Below, at and above the threshold

> These examples name the state at the condition, after the earlier update.

Identify what each test value refers to

Compare the three values immediately around the **threshold**. These are the values when the **condition** is reached, after the initial add-2 instruction.

Testing only a value far above 6 would not show whether equality was handled correctly.

Compare 5,6,7 at the condition

Three checks around the threshold

SCORE AT THE CONDITION	CHOOSE ONE BRANCH	SCORE AFTER THE BRANCH
Score is 5	At least 6 is false so add 4	Score becomes 9
Score is 6	At least 6 is true so subtract 3	Score becomes 3
Score is 7	At least 6 is true so subtract 3	Score becomes 4

At least includes equality. The middle row is essential for checking this boundary. Read each row as a separate run; do not pass the result of one row into the next.

Distinguish four comparison phrases

Above 6 is different: it means strictly greater than 6, so 6 itself does not pass. At most 6 means 6 or smaller. Below 6 means strictly smaller, so it excludes 6. Say the equality case aloud before using one of these phrases.

True does not mean larger or better

A **condition** can be true even when its chosen action makes the number smaller. True describes the test result; it does not mean good, bigger, or a successful game. The written instructions define the action.

Explain one branch

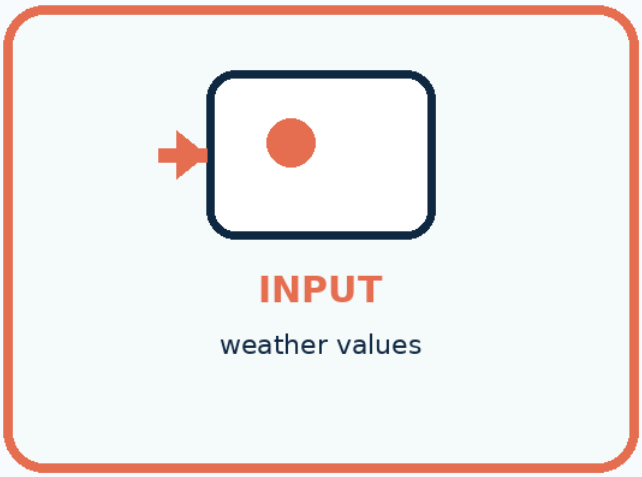
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

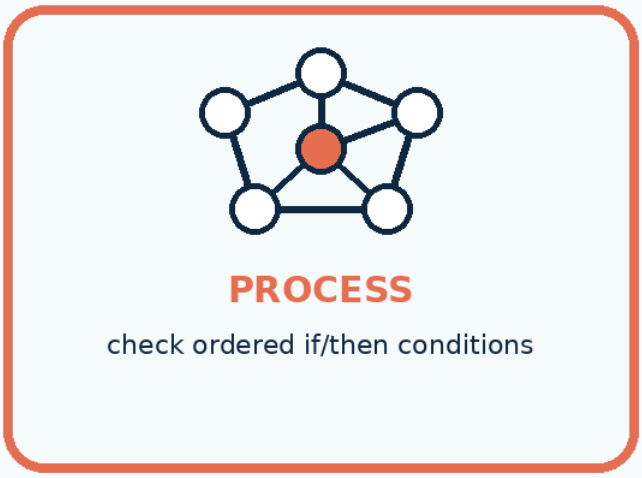
▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT weather values

PROCESS check ordered if/then **condition**s

OUTPUT clothing suggestion and reason

CORE IDEA **Condition**s make an algorithm **branch** when information changes.

UNDER THE HOOD A **condition**al evaluates a Boolean expression and selects a **branch**. Order matters when **condition**s overlap. Complete decision logic needs explicit handling for boundary values and an **else** path.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare conditional decisions

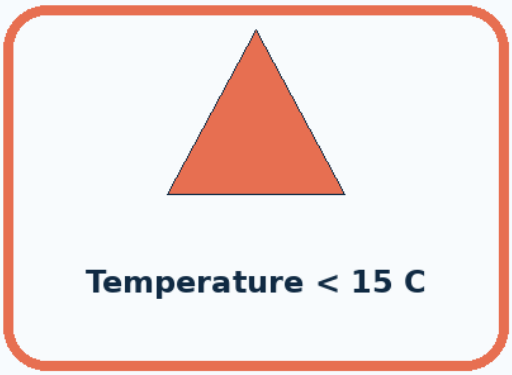
> Use the stated condition, equality rule and available input to justify the selected path.

Pictures paired with their evidence



The case to investigate

CASE 1 — Rain = yes Clue: then choose raincoat. Meaning: clear binary **branch**.



The case to investigate

CASE 2 — Temperature < 15 C Clue: then choose warm layer. Meaning: numeric **threshold**.



The case to investigate

CASE 3 — Exactly 15 C Clue: needs boundary rule. Meaning: edge case.



The case to investigate

CASE 4 — Rain and cold Clue: two **condition**s overlap. Meaning: order or combined rule needed.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY Never hide an unhandled case between two **threshold**s.

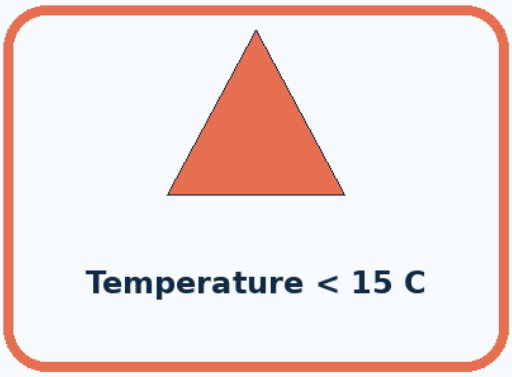
▼ **Earlier wording, alternatives and picture transcripts**

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Rain = yes Clue: then choose raincoat. Meaning: clear binary **branch**.

CASE 2 — Temperature < 15 C Clue: then choose warm layer. Meaning: numeric **threshold**.

CASE 3 — Exactly 15 C Clue: needs boundary rule. Meaning: edge case.

CASE 4 — Rain and cold Clue: two **condition**s overlap. Meaning: order or combined rule needed.

Change one thing in the pictured case

Change: Replace at least 6 with above 6.

Prediction: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

MAIN LESSON · SECTION 6

One updated value. One selected branch.

> This separate counter subtracts 1 when the updated value is at least 6, otherwise it doubles that value.

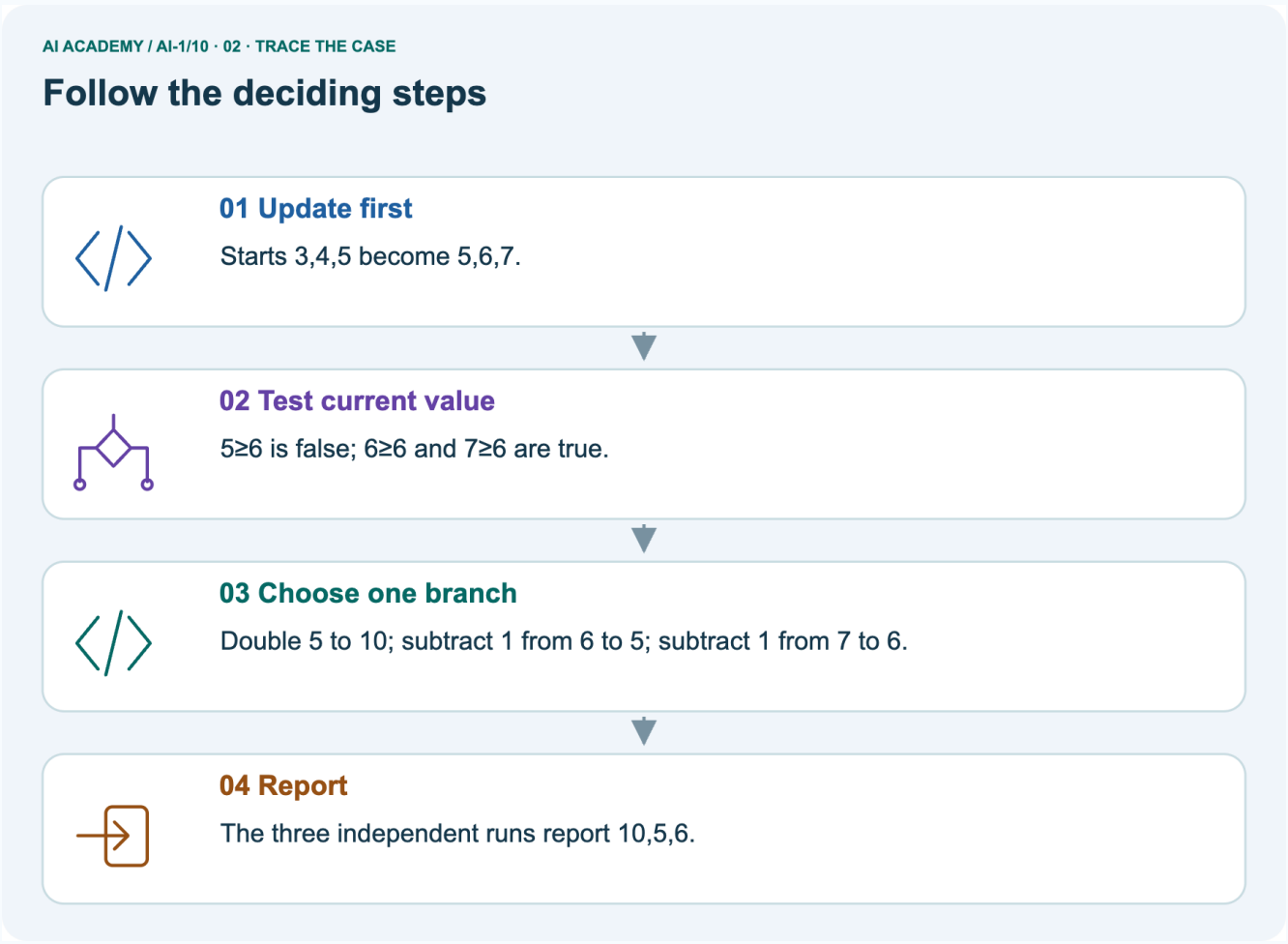
The counter’s exact recipe and three starts

A fictional counter begins with a number, adds 2, then chooses one **branch**.

If the updated value is at least 6, subtract 1; otherwise double it.

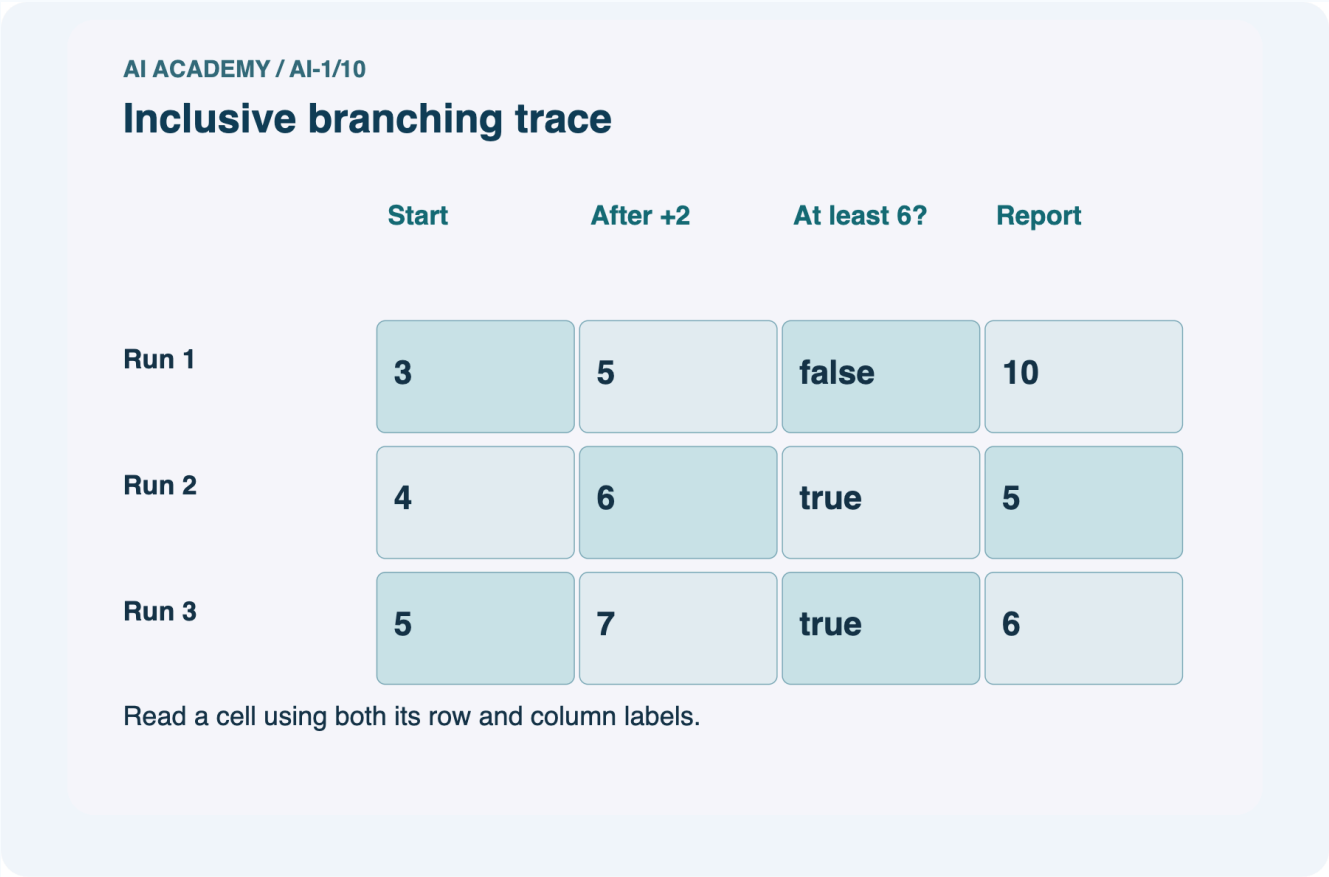
Report once and stop. Use separate starting values 3,4,5.

Trace the three separate runs



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the inclusive boundary results



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why the branch does not repeat

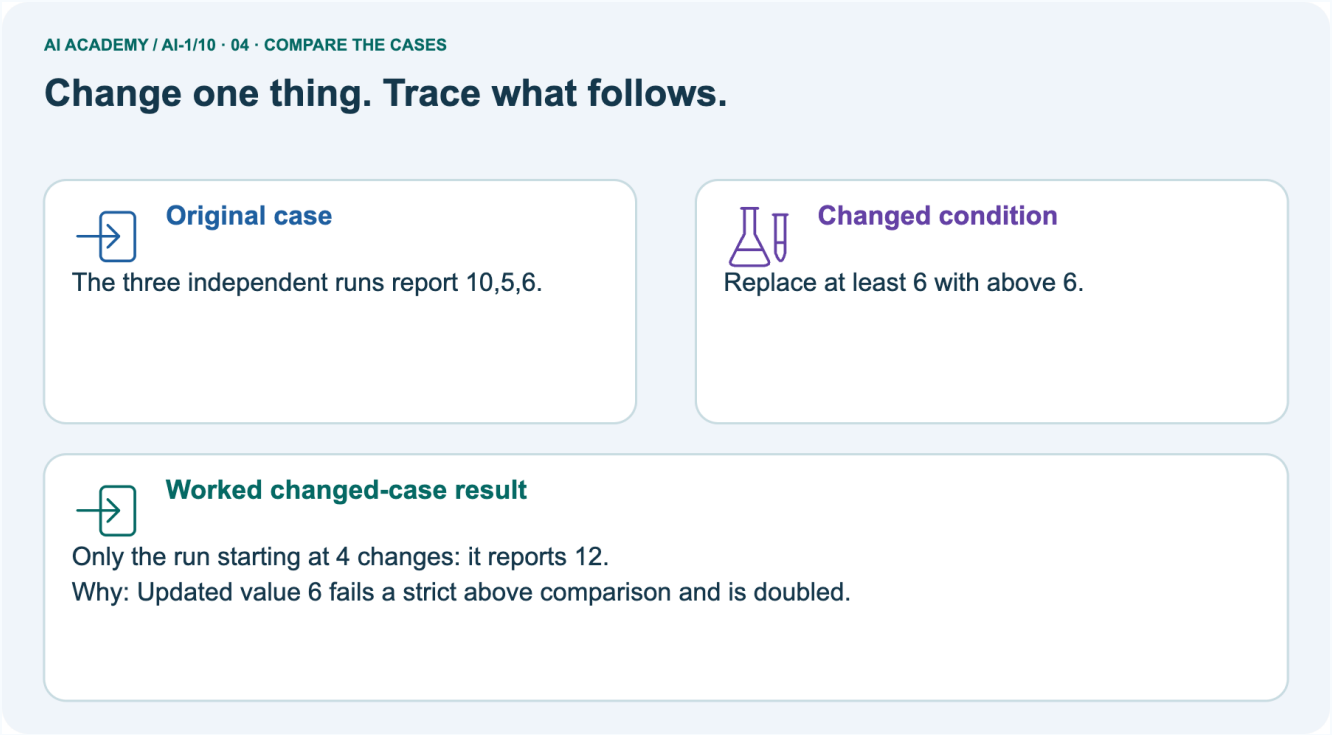
The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Predict a strict boundary instead

- Before checking, replace only “at least 6” with “above 6”. Use the counter’s subtract-1-or-double recipe, not the earlier paper game’s recipe. Trace starts 3,4,5 and name the single input whose result changes.

▼ Check the changed comparison



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit what a final number tells you

A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

Explain the distinguishing input

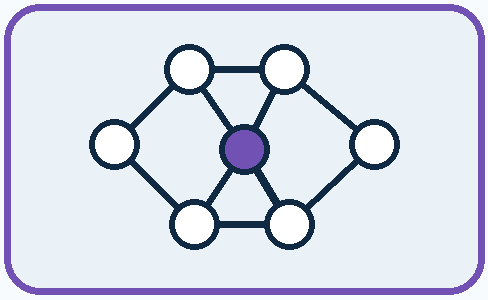
- Read the labelled picture: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE A clothing helper follows different paths depending on temperature, rain, and wind **condition**s.

STEP 1 — OBSERVE weather values

STEP 2 — TRACE check ordered if/then **condition**s

STEP 3 — CONCLUDE clothing suggestion and reason

MODEL REASONING Each test follows the stated **condition** order. The exactly-at-**threshold** case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined **output**s or a clear priority.

DO NOT OVERCLAIM Never hide an unhandled case between two **threshold**s.

What new **evidence** could change this answer?

A closer explanation

Initial score	State and condition after adding 2	Branch and final score
3	5 is not at least 6	Add 4 and report 9
4	6 is at least 6	Subtract 3 and report 3
5	7 is at least 6	Subtract 3 and report 4

In the first run, the false **branch** increases 5 to 9. That does not make the earlier decision run again. The procedure has already chosen and carried out the otherwise **branch**, so it moves on to report and stop.

Subtracting 3 afterward would incorrectly run both **branch**es and report 6.

This rule applies to one if/**else** decision. A later, separate **condition** would be a new test of the state at that later point. Do not invent an extra test or loop because the value crossed the **threshold**.

Follow only the instructions in the procedure.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: If/Then Decisions

The facts we will use

AI ACADEMY / AI-1/10

Inclusive branching trace

	Start	After +2	At least 6?	Report
Run 1	3	5	false	10
Run 2	4	6	true	5
Run 3	5	7	true	6

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/10

If/Then Decisions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Update first

Starts 3,4,5 become 5,6,7.

2. Test current value

5>=6 is false; 6>=6 and 7>=6 are true.

3. Choose one branch

Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.

4. Report

The three independent runs report 10,5,6.

CHANGE ONE THING

Replace at least 6 with above 6.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Update first	Starts 3,4,5 become 5,6,7.
2. Test current value	5≥6 is false; 6≥6 and 7≥6 are true.
3. Choose one branch	Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6.
4. Report	The three independent runs report 10,5,6.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: Replace at least 6 with above 6. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Unknown is not automatically false

> A missing value needs its stated fallback before the procedure selects a branch.

Handle an unreadable score explicitly

Our examples so far use known whole-number scores. If score is unreadable when the test is reached, we cannot establish whether it is at least 6. Unknown does not mean false, and a blank is not automatically zero.

For this paper exercise, pause the procedure and request the missing value before selecting a **branch**.

Keep boundary tests separate from missing-input handling

A complete procedure must specify what to do when required **input** is unavailable. Another system might define a separate fallback, but it must be written explicitly.

Testing below, at and above the boundary checks comparison behaviour; it does not replace a missing-**input** policy or prove every possible implementation correct.

Design the equality test

- CLAIM A If one example works, the **branch** is complete.
- CLAIM B Changing **branch** order cannot change the result.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A tester checks only scores 2 and 10 at the condition and says the at-least-6 comparison is correct. Which additional value is most useful for detecting an accidental above-6 comparison, and why**

▼ Earlier wording, alternatives and picture transcripts

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Rain = yes	then choose raincoat	clear binary branch
Temperature < 15 C	then choose warm layer	numeric threshold
Exactly 15 C	needs boundary rule	edge case
Rain and cold	two conditions overlap	order or combined rule needed

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: expose the equality mistake

> Use a start that reaches the threshold after the update.

Compare the paper game’s intended and copied conditions

Suppose a copied procedure says above 6 instead of at least 6. A score of 5 at the **condition** takes the false **branch** under both versions. A score of 7 takes the true **branch** under both.

But exactly 6 separates them: the intended version subtracts 3 and gives 3; the copied version adds 4 and gives 10.

> **Choose an initial score that reaches exactly 6 after the first add-2 instruction. Trace both the intended at-least version and the copied above version. Explain why this test reveals an error that the neighbouring values can miss**

Keep condition and update evidence separate

Record the initial value, the value at the **condition**, the true or false result, and the chosen update. A final number alone may not reveal whether the test or the arithmetic was wrong.

Keep the expected behaviour tied to the stated policy instead of changing it to match a result.

Use W2 to read all four boundary phrases

Use workbook W2 for the connected example "**Condition**s and boundary cases". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “the condition reads the start”

- Use the worked case for **Condition**s and boundary cases. Proposed explanation: “The **condition** reads the starting value”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the pictured counter

Which **input** detects the difference between the two written **condition**s? Explain its entire run.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: **Condition**s make an algorithm **branch** when information changes.

Use a fresh example to explain **condition**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Changing **branch** order cannot change the result.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Read four boundary phrases

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate four weather-rule cases

> Use a complete policy with an equality case and an explicit missing-input result.

The stated policy and stopping points

Use fictional wardrobe-choice cards with rain=yes/no and a temperature in degrees C. Our paper policy: if either required input is unknown, report REVIEW and stop. Otherwise, if rain=yes, choose RAINCOAT; else if temperature is below 18, choose JUMPER; otherwise choose LIGHT LAYER. Report that one choice and stop. This is a declared paper policy; a person decides what is appropriate in a real situation.

Predict, trace and compare

1. Predict and trace four separate cards: rain=yes,20 C; rain=no,17 C; rain=no,18 C; rain=unknown,20 C. Record the available inputs, each test reached, the branch and the final label. Do not run the temperature branch after choosing RAINCOAT. Have a partner trace the same cards and inspect the first disagreement.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep the branch evidence

Case	Prediction	Observed result	What it means
Normal			
Edge			

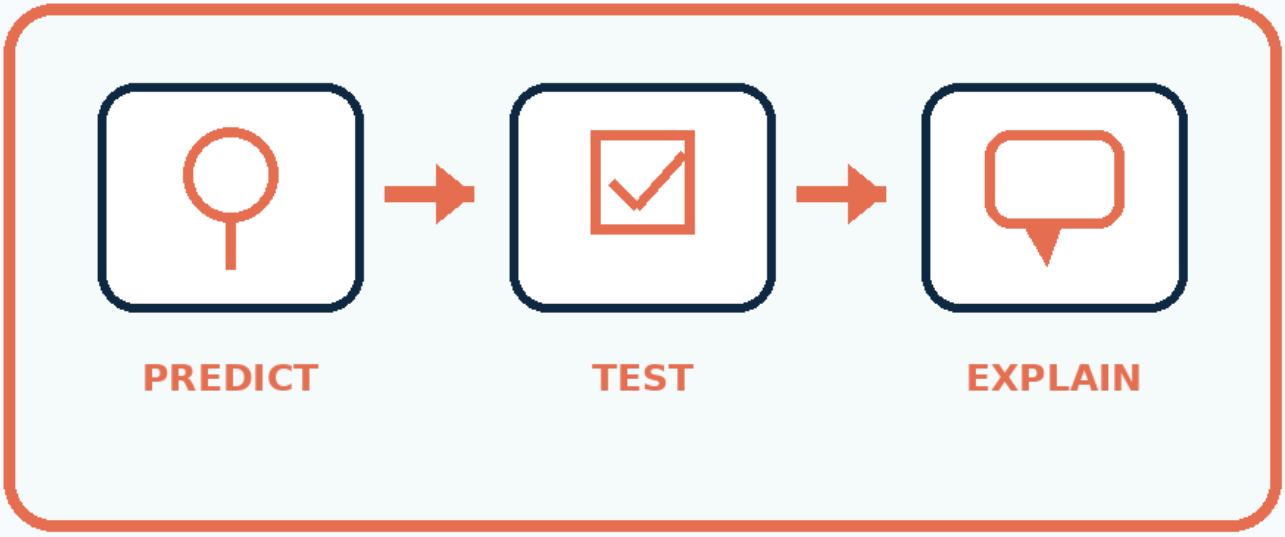
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EXPLORER LAB

Weather Wardrobe Branches

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Build a weather-clothing rule and test four cases.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: transfer the method to brightness

> Use the new add-3 update, at-most-7 comparison and its stated branch actions.

The new procedure

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Transfer the method to a new display

Week 10 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Add a transparent branch to your sorter

> Include the equality case, a complete else path and an unavailable-input action.

This week’s project decision

THIS WEEK'S CHECKPOINT Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

Plan the information, tests and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?

- What happens inside the **process**?
- What normal and edge cases will you test?
- When should it say not sure or ask a person?

Check for gaps between thresholds

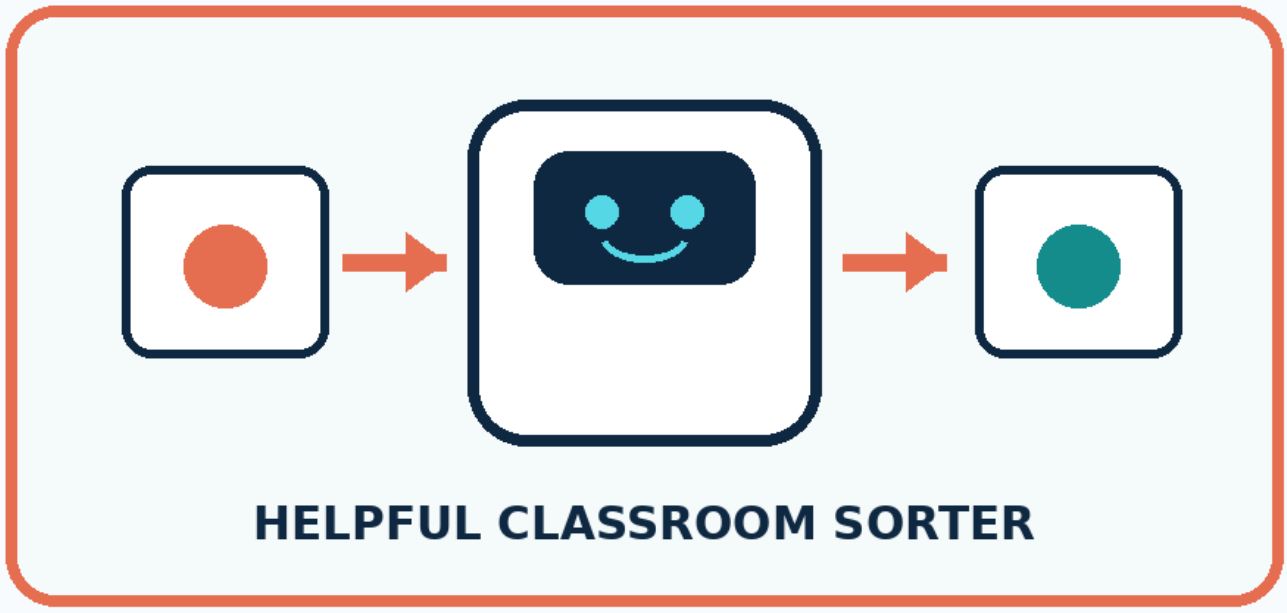
RESPONSIBLE DESIGN Never hide an unhandled case between two **threshold**s.

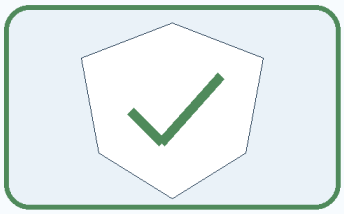
▼ **Earlier wording, alternatives and picture transcripts**

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify the equality branch

> Close your notes and name the state at the test, its result and the one action selected.

Your exit explanation

- Close your notes. Explain this goal in your own words: Evaluate a **condition** using the current state and trace only the selected **branch**

My deciding step, evidence and one remaining question

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
condition	question evaluated as true or false	
branch	path selected by a condition	
threshold	value where a decision changes	
else	path used when earlier conditions are false	

Terms for the connected explanation

Word	Meaning
Inclusive boundary	A boundary whose equality case is included
Boundary test	A test using a value at or close to a point where behaviour can change

Repair a claim using a word

- CLAIM TO REPAIR If one example works, the **branch** is complete.
- Use one vocabulary word and one piece of **evidence** to repair it:

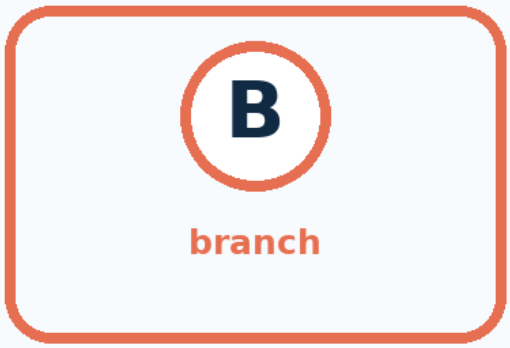
▼ **Earlier wording, alternatives and picture transcripts**

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

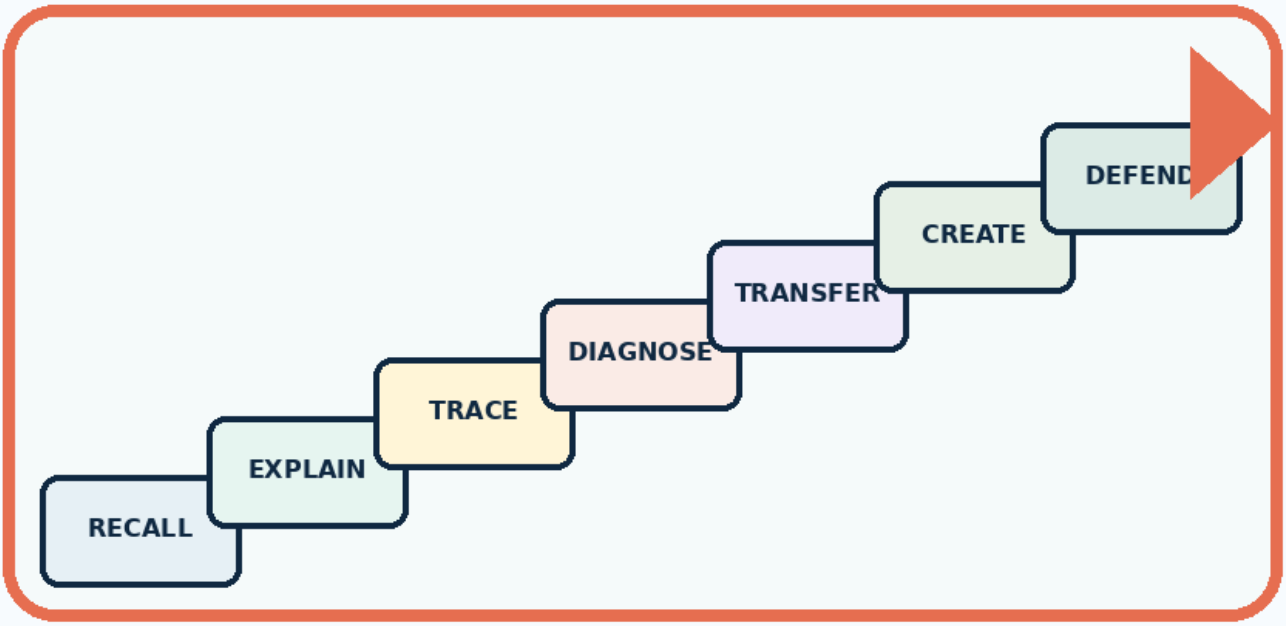
- RECALL Define **condition**.
- EXPLAIN **Condition**s make an algorithm **branch** when information changes.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: If one example works, the **branch** is complete.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Transfer the method to a new display is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

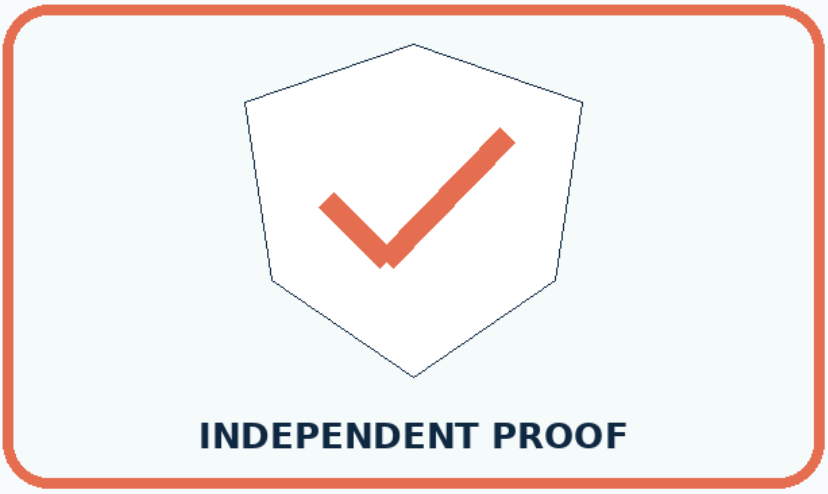
My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ Earlier wording, alternatives and picture transcripts

MASTERY

Checklist for explaining this week's idea

Use after practice. Transfer the method to a new display is the selected independent case.



EXPLANATION CHECK Record the state at the condition, test equality explicitly, follow only the selected branch and preserve uncertainty when an input is missing.

Independent **evidence**:

SELF-CHECK Each test follows the stated **condition** order. The exactly-at-**threshold** case reveals whether the rule uses less than or less-than-or-equal. Rain-plus-cold needs combined **output**s or a clear priority.

NEXT BRIDGE Carry today's **evidence** habit into: Repeating Efficiently.

▼ **Extra practice: Does this card have a star?**

REFERENCE · SECTION 13

Extra practice: Does this card have a star?

> Try the question before opening its worked explanation.

Read the supplied case and question

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Your explanation

- Which detail supports the answer, and why does it matter?

> **Write or draw your reasoning and evidence.**

▼ Check after trying

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

Evidence or stage	Value or result
Star only	Yes → tray A
Circle only	No → tray B
Star and circle	Yes → tray A

▼ Earlier wording, alternatives and picture transcripts

Week 10 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Does this card have a star?

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Star and mixed cards go to A because both have a star. The circle goes to B. The test is has a star, not has only a star.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Evaluate a condition using the current state and trace only the selected branch	Explain the deciding step, show a trace or test, and state one limit.
Distinguish inclusive and strict boundaries using values below, at and above a threshold	Explain the deciding step, show a trace or test, and state one limit.
Design a boundary test and handle an unknown input without silently treating it as false	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional counter begins with a number, adds 2, then chooses one **branch**.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional counter begins with a number, adds 2, then chooses one **branch**.
- Which labelled step matches this detail? The three independent runs report 10,5,6.
- 1. Update first
- 2. Report
- 3. Choose one **branch**
- 4. Test current value
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Report The three independent runs report 10,5,6. The **condition** is evaluated after adding 2.

Equality belongs to the at-least **branch**, so starting at 4 checks the boundary. Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Changed-case question

- Changed-case practice: Replace at least 6 with above 6. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. Only the run starting at 4 changes: it reports 12.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: Only the run starting at 4 changes: it reports 12. Updated value 6 fails a strict above comparison and is doubled.

Repair and save your work

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

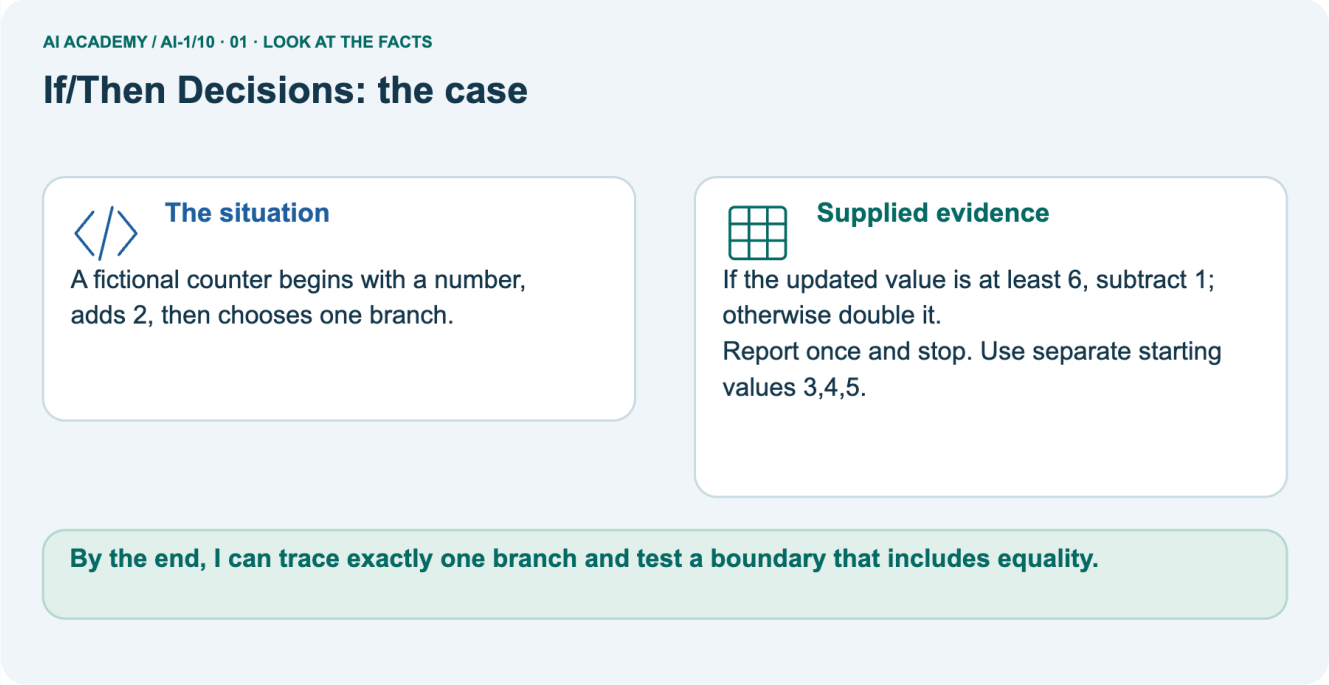
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. If/Then Decisions: the case

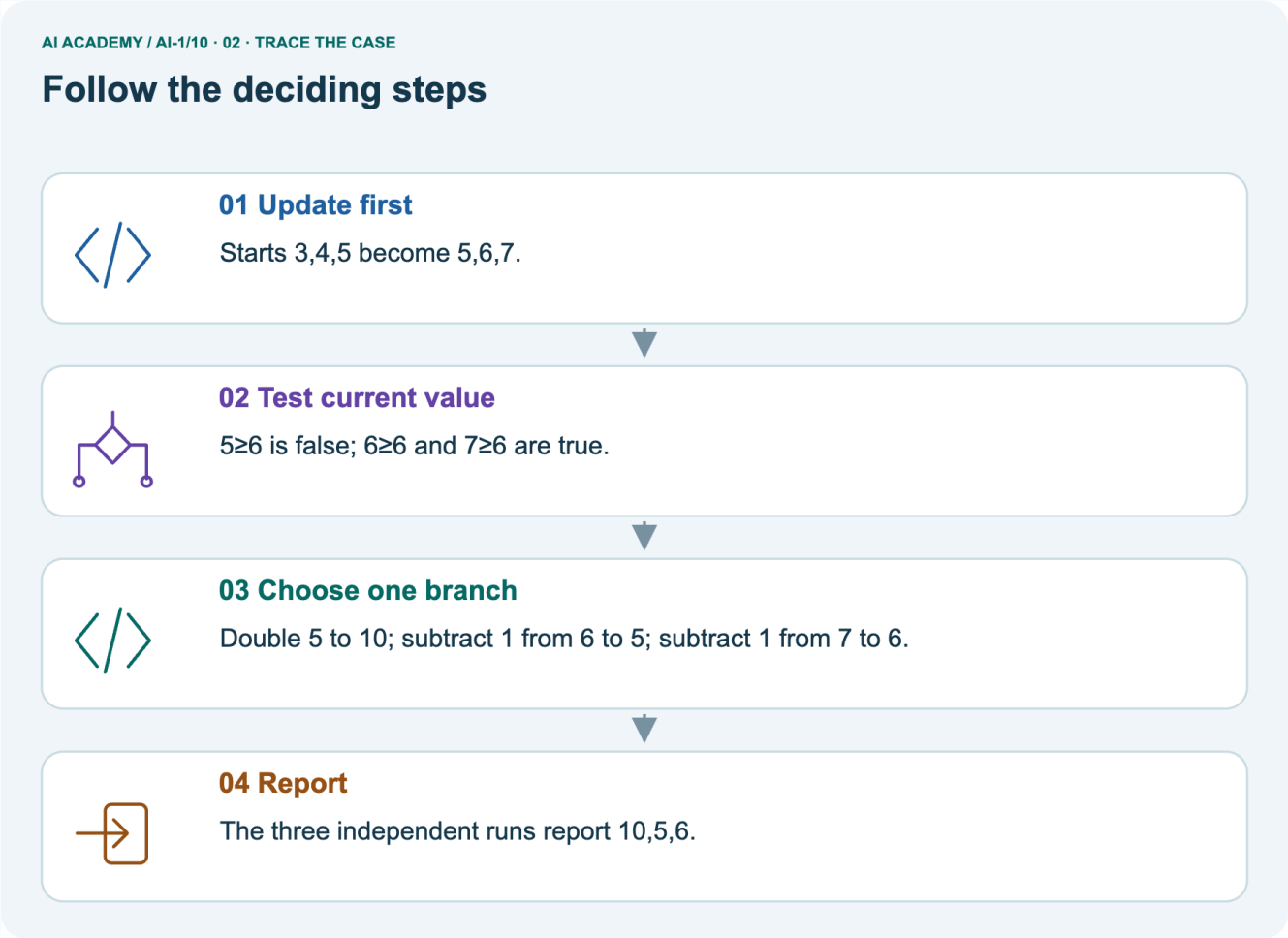


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter begins with a number, adds 2, then chooses one **branch**. Supplied facts: If the updated value is at least 6, subtract 1; otherwise double it. Report once and stop. Use separate starting values 3,4,5.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

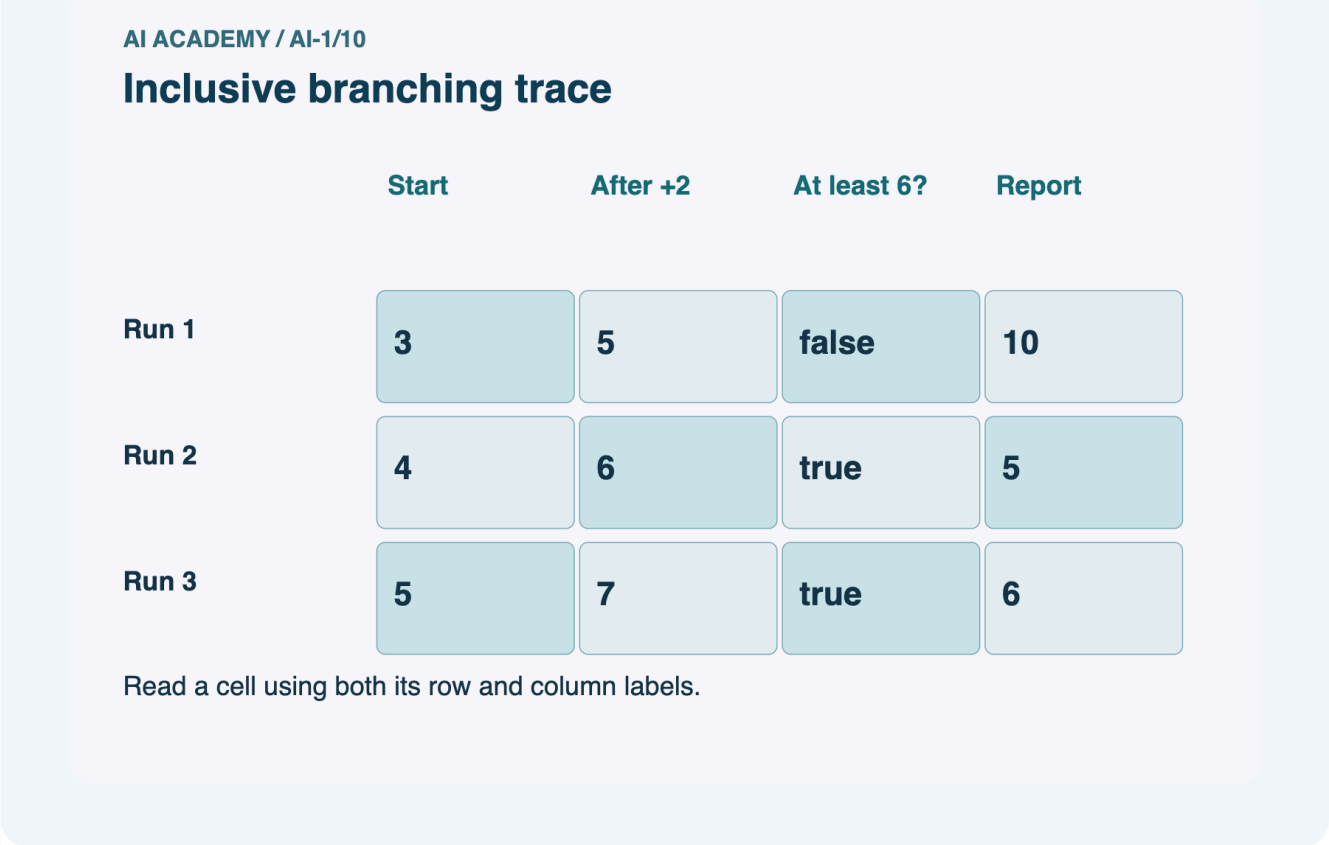


Follow the labelled picture with the matching explanation. Open the vector diagram

Update first: Starts 3,4,5 become 5,6,7. → Test current value: 5≥6 is false; 6≥6 and 7≥6 are true. → Choose one **branch**: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. → Report: The three independent runs report 10,5,6.

Point to each numbered stage and explain which recorded **input** it uses.

03. Inclusive branching trace

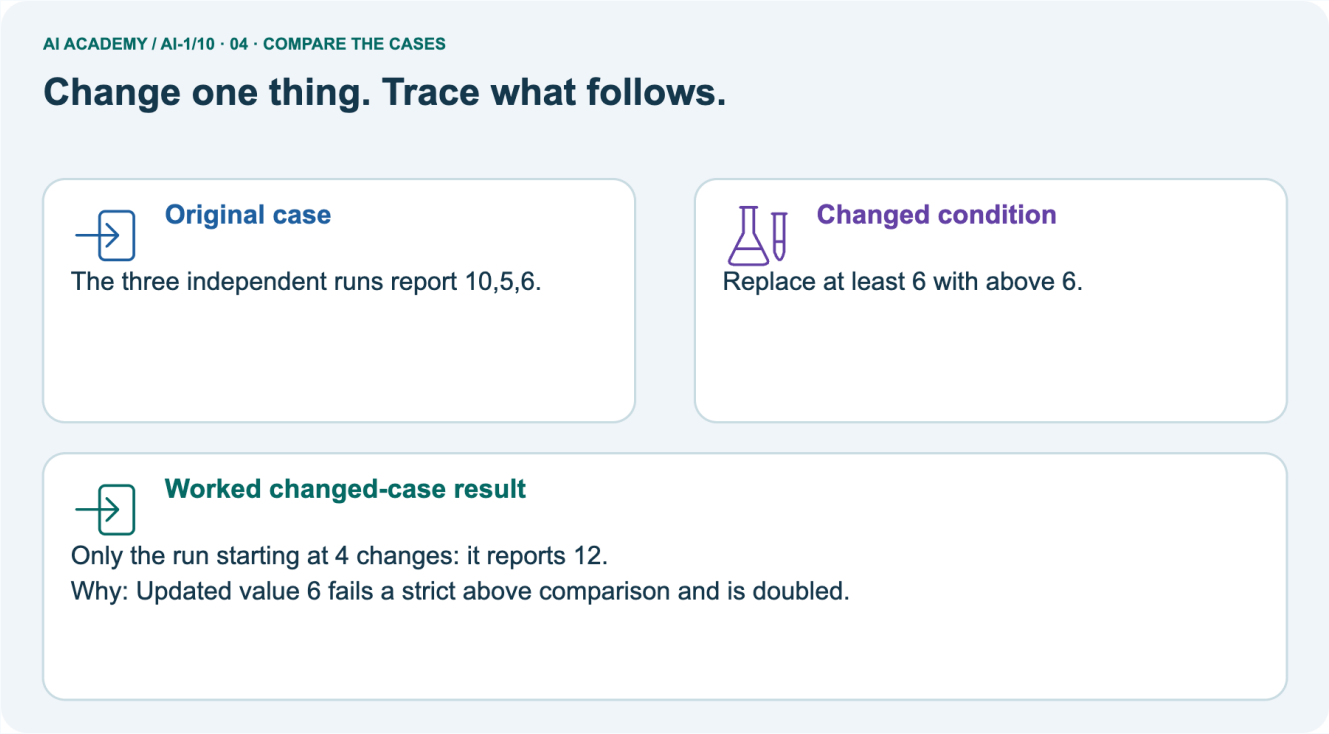


Follow the labelled picture with the matching explanation. Open the vector diagram

Inclusive **branch**ing trace. Run 1: Start = 3, After +2 = 5, At least 6? = false, Report = 10. Run 2: Start = 4, After +2 = 6, At least 6? = true, Report = 5. Run 3: Start = 5, After +2 = 7, At least 6? = true, Report = 6.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



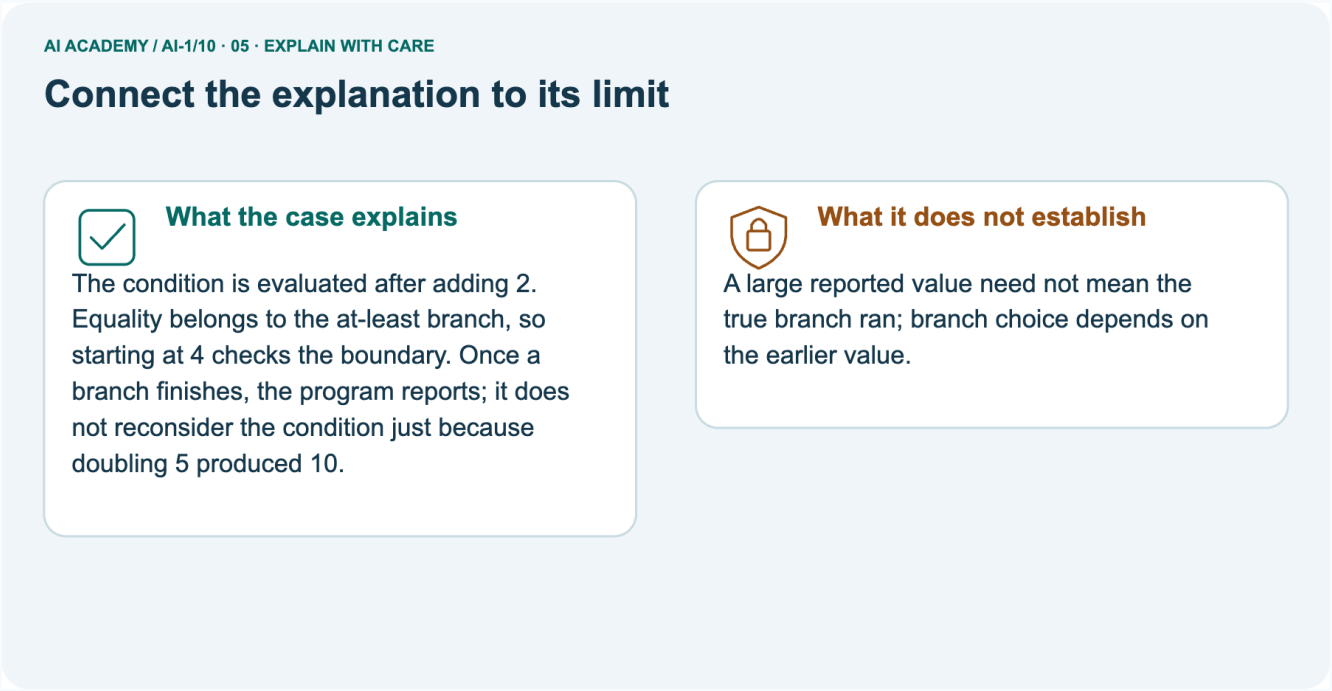
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The three independent runs report 10,5,6. Changed **condition**: Replace at least 6 with above 6. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

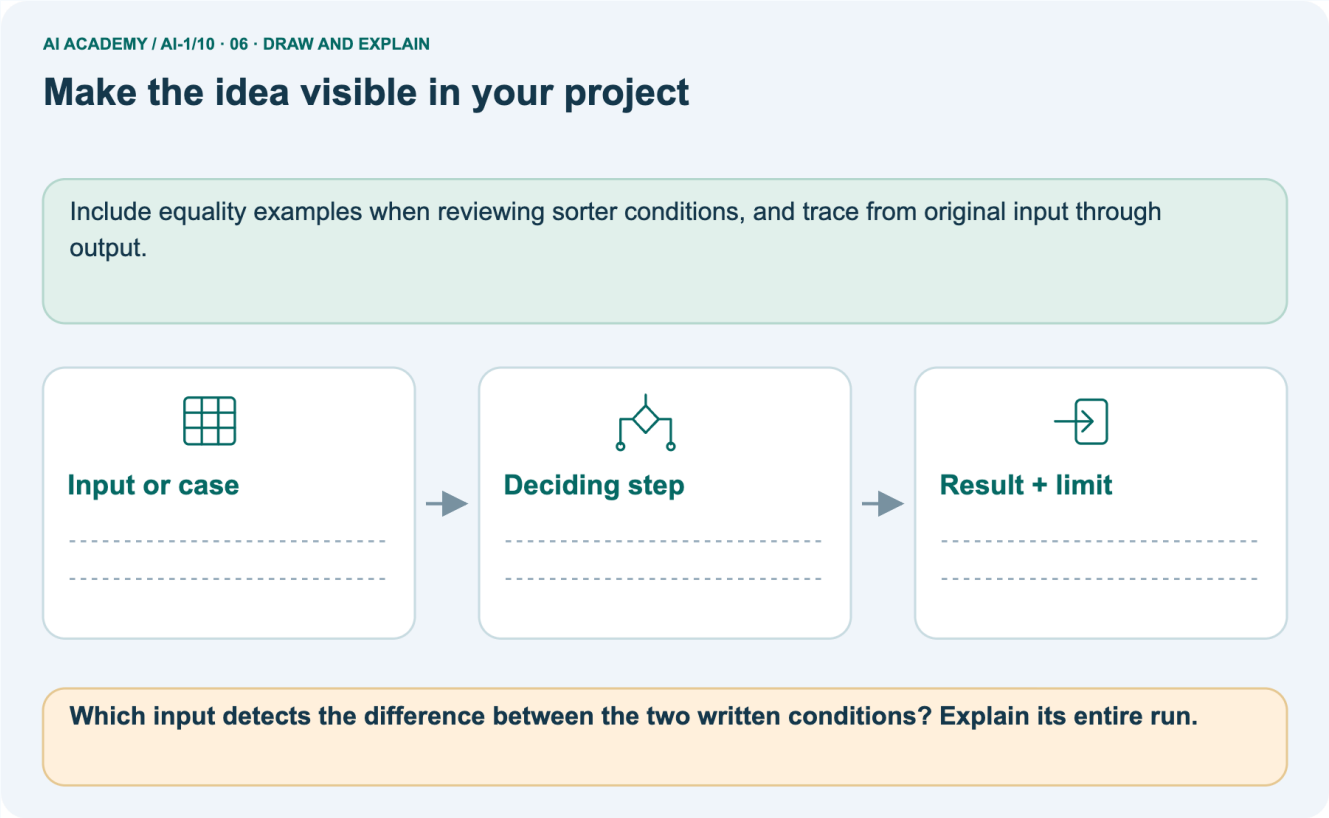
The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Limit: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. If/Then Decisions: the case

02. Follow the deciding steps

03. Inclusive branching trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.